5.Charring test:

When carbohydrates are treated with concentrated sulphuric acid, dehydration of carbohydrates results in charring.

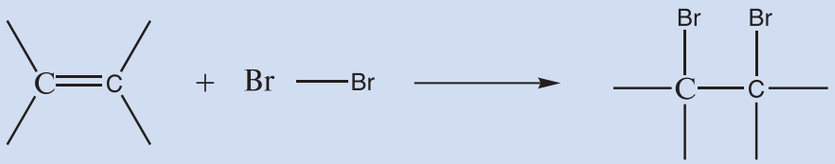
$$C_x(H_2O)_y \xrightarrow[\Delta]{H_2SO_4} x\,C + yH_2O$$

6. Ignition test

Aromatic compounds burn with a strong sooty yellow flame because of the high carbon–hydrogen ratio. Aliphatic compounds burn with non-sooty flame.

7.Test with bromine water:

In this test, the orange-red colour of bromine solution disappears when it is added to an unsaturated organic compound.

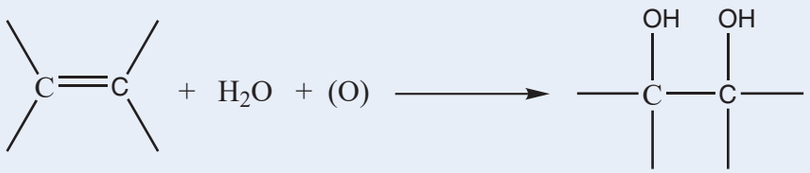


unsaturated compound — Orange-yellow — Colour less

8. Test with $KMnO_4$ (Baeyer's Test)

In this test, pink colour of $KMnO_4$ disappears, when alkaline $KMnO_4$ is added to an unsaturated hydrocarbon. The disappearance of pink colour may take place with or without the formation of brown precipitate of MnO_2.

$$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3(O)$$



unsaturated compound — Colour less

9. Neutral $FeCl_3$ test:

Phenol reacts with ferric ions to form violet coloured complex.

Aqueous solution of naphthols do not give any characteristic colour with neutral ferric chloride. But alcoholic solution of α and β naphthols give blue-violet and green colouration respectively due to the formation of binaphthols.

organic analysis.indd 261 22-12-2022 19:48:33